Sets 2 — Practical problems · Three sets · Shading · Set-builder notation

Marks in brackets. ★ = harder question. $\mathscr{E}$ is the universal set; A' is the complement of A.

1 In a class of 30 students, 18 study French, 15 study Spanish and 5 study neither language. [5]

a Draw a Venn diagram, using x for the number who study both. **b** Find x.

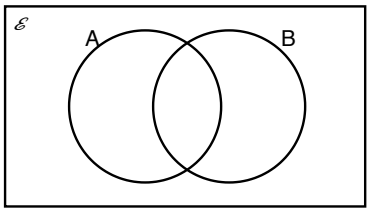
c How many students study exactly one of the two languages?

a Two overlapping circles F and S inside $\mathscr{E}$: x in the overlap, $18 - x$ in F only, $15 - x$ in S only, 5 outside both.

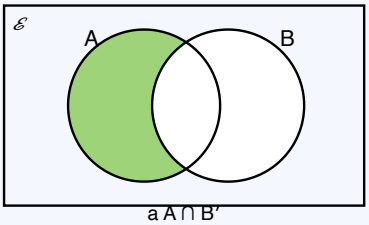
b The regions add up to 30:
$(18 - x) + x + (15 - x) + 5 = 30 \Rightarrow 38 - x = 30 \Rightarrow x = 8$.

c F only $= 10$, S only $= 7$: 17 students.

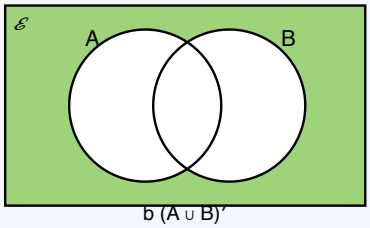
2 On two copies of the diagram, shade the sets **a** $A \cap B'$ and **b** $(A \cup B)'$. [4]



a $A \cap B'$ is "in A and not in B": the part of A outside the overlap. **b** $(A \cup B)'$ is "not in A or B": everything outside both circles.

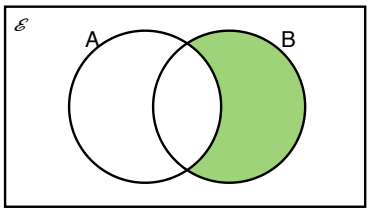


a A ∩ B′



b (A ∪ B)′

3 Describe the shaded set using set notation. [3]



The shaded region is inside B but outside A: $A' \cap B$ (also written $B \cap A'$).

4 $C = \{x : \ -3 \leq x < 3,\ x \in \mathbb{Z}\}$. List the members of C and write down $n(C)$. [3]

> Integers from -3 up to but not including 3: $C = \{-3, -2, -1, 0, 1, 2\}$, so $n(C) = 6$.

5 Write in set-builder notation the set of positive multiples of 7. [2]

> $\{x : \ x = 7n,\ n \in \mathbb{N}\}$, i.e. $\{7, 14, 21, \ldots\}$.

6 In a sports club of 120 members, 50 play basketball, 60 play volleyball and 45 play handball. 15 play basketball and volleyball, 20 play volleyball and handball, 12 play basketball and handball, and 6 members play none of the three sports. Use a Venn diagram to find how many members [8]

a play all three sports **b** play only volleyball.

> Let x play all three and fill the diagram from the middle outwards. The three "pairs" regions are $15 - x$ (B∩V only), $20 - x$ (V∩H only), $12 - x$ (B∩H only). Then B only
> $= 50 - (15 - x) - (12 - x) - x = 23 + x$, V only
> $= 60 - (15 - x) - (20 - x) - x = 25 + x$, H only
> $= 45 - (20 - x) - (12 - x) - x = 13 + x$.
>
> **a** All seven regions plus the 6 outside make 120:
> $(23 + x) + (25 + x) + (13 + x) + (15 - x) + (20 - x) + (12 - x) + x + 6 = 120$
> , which simplifies to $114 + x = 120$, so $x = 6$.
>
> **b** Only volleyball $= 25 + x = 31$. (Check: B only 29, V only 31, H only 19, pairs 9, 14, 6, all three 6 — total 114, plus 6 none $= 120$ ✓.)

Total: 25 marks

Check: put x in the overlap and work outwards, then add every region to the total · "neither" goes outside the circles · $A \cap B'$ = in A but not B; $(A \cup B)'$ = outside both · $\mathbb{N}$ natural numbers, $\mathbb{Z}$ integers, $\mathbb{Q}$ rationals, $\mathbb{R}$ reals · $\{x : \ \text{condition},\ x \in \text{set}\}$ reads "the set of x such that…".